

**23CSP501 MERN Stack Development
Assignment**

ONLINE FOOD ORDERING WEBSITE

Submitted by

Roll Number : 71782P122

Name : JAYA PRAKASH S

Department of Computer Science and Engineering

KARPAGAM COLLEGE OF ENGINEERING

(Autonomous)

Semester : V Academic

Year : 2025-26

1. Abstract / Executive Summary

The **Online Food Delivery System** is a MERN stack web application that enables customers to browse menus, add food items to their cart, and place secure online orders. It includes features like authentication (email/Google login), profile management, cart system, and order tracking. The frontend is built with **React.js** and reusable components (Navbar, Header, Food Display, Cart, PlaceOrder, etc.), while the backend uses **Node.js/Express.js** with **MongoDB Atlas** for storing users, food items, and orders. The platform ensures a smooth experience for customers and provides restaurants with efficient order and menu management.

2. Introduction

2.1 Problem Statement

Efficient food ordering and delivery is a growing need in today's fast-paced lifestyle. Traditional systems often lack convenience, personalization, and real-time tracking. The **Online Food Delivery System** overcomes these challenges by offering a structured MERN-based solution where customers can browse menus, add items to their cart, and place secure orders, while restaurants can manage food items and monitor sales. The platform ensures smooth navigation, authentication, and real-time updates, creating a seamless experience for both customers and service providers.

3. Requirements

3.1 Functional Requirements

- **FR-1:** ☐ User Authentication (Login/Signup + Google OAuth)
- **FR-2:** Browse food categories and items (via FoodDisplay & ExploreMenu components)
- **FR-3:** Add, update, and remove items from Cart
- **FR-4:** Place orders and store order details in database
- **FR-5:** Order tracking with status updates (pending, confirmed, delivered)
- **FR-6:** User dashboard to view past orders and profile details
- **FR-7:** Admin/Restaurant dashboard to add, update, and manage food items
- **FR-8:** Responsive Navbar, Header, Footer, and Sidebar navigation
- **FR-9:** Promotional/celebration pages (e.g., offers, special deals)
- **FR-10:** Contact form submission to backend
- **FR-11:** Profile picture upload and retrieval from MongoDB
- **FR-12:** Toasts and alerts for user feedback (success, error, order confirmation)

4. System Design & Architecture

4.1 High-Level Architecture

- **Frontend:** React + React Router (UI, routing, session handling)
- **Backend API:** Node.js/Express (auth, employees, tasks, verification, contact messages)
- **Database:** MongoDB Atlas with three main collections (TaskList, EmplTask, VerifyTask)

4.2 Technology Stack

- **Frontend:** React 18, Material-UI (MUI)
- **Routing:** React Router DOM
- **Auth:** UserId/password login
- **State Management:** React Hooks + localStorage
- **Backend:** Node.js, Express.js
- **Database:** MongoDB Atlas
- **Hosting:** Netlify (frontend), Render/Heroku (backend)

4.3 Component Design

Component	Responsibility
Header / Navbar	Displays branding (logo, app name) and profile/avatar menu with navigation links.
Footer	Footer Contains navigation links, contact details, and social media info.
Login / Signup	Login / Signup Handles user authentication (email/password, Google OAuth) and stores session.
Home / Dashboard	Home / Dashboard Displays food categories, recommended items, and user details (orders, profile).
Food Item Display	Food Item Display Browse available food items by category, view details, add/remove items from cart.
Cart	Cart Shows selected items, quantities, total price, and checkout option.
Place Order	Place Order Captures delivery details, confirms orders, and stores them in database.
Order Tracking	Order Tracking Displays order status (pending, confirmed, delivered) for users.
Admin Panel	Admin Panel Manage food items (add/update/delete) and view order history.
Promotions / Deals	Promotions / Deals Celebration or offer pages highlighting discounts and special menus.

4.4 Routing

Route	Page
/	Home (food categories, featured items)
/login	Login page
/register	Signup page
/cart	Cart – view, update, and checkout items
/place-order	Place Order – delivery details & order confirmation
/orders	User dashboard – view order history & tracking
/orders/:id	Order details (status: pending, confirmed, delivered)
/admin	Admin panel – manage food items & view orders

State Management Strategy

- Local state with React Hooks
- Persistent state in localStorage
- Axios for backend communication

5. Implementation

5.1 Project Setup

- ☐ **Frontend:** React (Vite) + MUI
- ☐ **Backend:** Node.js/Express
- ☐ **Database:** MongoDB Atlas

Setup Commands:

Frontend

```
npm create vite@latest emp-mgmt-frontend
```

```
cd emp-mgmt-frontend
```

```
npm install
```

```
npm run dev
```

Backend

```
mkdir emp-mgmt-backend
```

```
cd emp-mgmt-backend
```

```
npm init -y
```

```
npm install express mongoose bcryptjs jsonwebtoken cors dotenv multer
```

Directory Structure:

src/

assets/

components/

(Navbar, Header, Footer, ExploreMenu, FoodDisplay, FoodItem, Cart, PlaceOrder, Celebratie, LoginPop, Toasts)

css/

(Home.css, FoodDisplay.css, FoodItem.css, Cart.css, PlaceOrder.css, Celebratie.css, Login.css, Navbar.css,

Footer.css)

pages/

(Home, Cart, PlaceOrder, Orders, Celebratie, About, Terms, Contact, Login, Signup)

App.jsx

main.jsx

models/

(User.js, FoodItem.js, Order.js, Cart.js)

routes/

(auth.js, food.js, order.js, cart.js, contact.js)

middlewares/
 (authMiddleware.js, uploadMiddleware.js)
server.js

5.2 Environment Configuration

VITE_API_BASE_URL
VITE_OAUTH_CLIENT_ID
MONGO_URI
JWT_SECRET

5.2 CODE

Header.jsx

```
import React from "react";
import "./Header.css";
import { Link, useNavigate } from
"react-router-dom";
const Header = () => {
  const navigation = useNavigate();
  return (
    <div className="header">
      <div className="header-
contents">
        <h2>Order your favourite food
here </h2>
        <a href="#explore-menu">
          {" "}
          <button> View Menu</button>
        </a>
      </div>
    </div>
  );
};

export default Header;
```

Login.jsx

```
import React, { useState, useEffect } from "react";
import "./LoginPop.css";
import { assets } from "../../assets/assets";
import axios from "axios";
```

```
const LoginPop = ({ setshowlogin }) => {
  const [currentstate, setcurrentstate] = useState("Log in");
  const [uname, setName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [loggedInUser, setLoggedInUser] = useState(null);
  const [alertMessage, setAlertMessage] = useState("");

  useEffect(() => {
    const storedUser = localStorage.getItem("user");
    setLoggedInUser(storedUser ? JSON.parse(storedUser) : null);
  }, []);

  const updateUserState = (user) => {
    setLoggedInUser(user);
    localStorage.setItem("user", JSON.stringify(user));
    window.dispatchEvent(new Event("storage")); // 🚫 Important: Trigger update globally
  };

  const handleSubmit = async (e) => {
    e.preventDefault();

    const endpoint = currentstate === "Log in" ? "login" : "Authentication";
    const data = currentstate === "Log in" ? { email, password } : { uname, email, password };

    try {
      const response = await axios.post(`http://localhost:3001/${endpoint}`, data);
      console.log("Response:", response.data);

      if (response.data.status === "Success") {
        setAlertMessage(✅ `${currentstate === "Log in" ? "Login" : "Sign Up"} successful! 🎉`);

        if (currentstate === "Log in") {
          const userData = { uname: response.data.uname, email };
          updateUserState(userData);
        }

        setTimeout(() => {
          setAlertMessage("");
          setshowlogin(false);
          setName("");
          setEmail("");
        }, 2000);
      }
    } catch (error) {
      console.log("Error:", error);
    }
  };
}
```

```
    setPassword("");
  }, 2000);
} else {
  setAlertMessage(' ❌ ${response.data.message} || "An error occurred."' );
}
} catch (error) {
  console.error("Error:", error);
  setAlertMessage(" ❌ Server error. Please try again.");
}
};

const handleLogout = () => {
  localStorage.removeItem("user");
  setLoggedInUser(null);
  window.dispatchEvent(new Event("storage")); // 🟡 Important: Trigger update globally
  setAlertMessage(" ✅ Logged out successfully! 🎉 ");

  setTimeout(() => {
    setAlertMessage("");
  }, 3000);
};

return (
  <div className="loginpop">
    {alertMessage} && <div className="blue-alert">{alertMessage}</div>

    <div className="login-pop-container">
      <div className="login-pop-title">
        <h2>{loggedInUser ? `Welcome, ${loggedInUser.uname}! 😊` : currentstate}</h2>
        <img onClick={() => setshowlogin(false)} src={assets.cross_icon} alt="Close" />
      </div>

      {loggedInUser ? (
        <div className="login-pop-logged-in">
          <button className="logout-btn" onClick={handleLogout}>Logout</button>
        </div>
      ) : (
        <form onSubmit={handleSubmit}>
          <div className="login-pop-inputs">
            {currentstate === "Sign Up" && (
              <input type="text" placeholder="Enter your name" required value={uname} onChange={(e) =>
                setName(e.target.value)} />
            )}
          </div>
        </form>
      )}
    </div>
  </div>
);
```

```
<input type="email" placeholder="Enter your email" required value={email} onChange={(e) =>
setEmail(e.target.value)} />
<input type="password" placeholder="Enter your password" required value={password}
onChange={(e) => setPassword(e.target.value)} />
</div>

<button>{currentstate === "Sign Up" ? "Create Account" : "Log In"}</button>

<p>
  {currentstate === "Log in" ? "Create a new account " : "Already have an account? "}
  <span onClick={() => setcurrentstate(currentstate === "Log in" ? "Sign Up" : "Log in")}>Click
here</span>
</p>
</form>
)}
</div>
</div>
);
};

export default LoginPop;
```

App.jsx

```
import React, { useState } from "react";
import Navbar from "../components/Navbar/Navbar";
import { Route, Routes } from "react-router-dom";

import Cart from "../pages/cart/cart";
import Home from "../pages/Home/Home";
import PlaceOrder from "../pages/placeorder/placeorder";
import Footer from "../components/Footer/Footer";
import LoginPop from "../components/LoginPop/LoginPop";
import Celebratie from "../pages/Celebratie/Celebratie";

const App = () => {
  const [showlogin, setshowlogin] = useState(false);
  return (
    <>
      {showlogin ? <LoginPop setshowlogin={setshowlogin} /> : <</>}
      <div className="app">
        <Navbar setshowlogin={setshowlogin} />
      </div>
    </>
  );
};
```

```
<Routes>
  <Route path="/" element={<Home />} />
  <Route path="/cart" element={<Cart />} />
  <Route path="/order" element={<PlaceOrder />} />
  <Route path="/celebratie" element={<Celebratie />} />
</Routes>
</div>
<Footer />
</>
);
};

export default App;
```

ExploreMenu.jsx

```
import React, { useEffect, useRef } from "react";
import "./ExploreMenu.css";
import { menu_list } from "../assets/assets";

const ExploreMenu = ({ category, setcategory }) => {
  const scrollRef = useRef(null);

  useEffect(() => {
    const scroll = () => {
      if (scrollRef.current) {
        if (scrollRef.current.scrollLeft >= scrollRef.current.scrollWidth / 2) {
          scrollRef.current.scrollLeft = 0; // Reset scroll to the beginning
        } else {
          scrollRef.current.scrollLeft += 1; // Move scroll position
        }
      }
    }
  });

  const interval = setInterval(scroll, 20); // Adjust speed if needed
  return () => clearInterval(interval);
}, []);

return (
  <div className="explore-menu" id="explore-menu">
    <h1>Explore Your Favourite Dish with Us</h1>
    <p className="explore-menu-text">
      Explore a world of flavors with our food ordering app! From sizzling
```

starters to indulgent desserts, our curated menu has something for everyone. Choose from fresh salads, hearty mains, and international cuisines, all crafted to perfection. Craving comfort food or trying something new? Dive into our mouthwatering selection now!

</p>

```
<div className="explore-menu-container" ref={scrollRef}>
  <div className="explore-menu-list">
    {menu_list.concat(menu_list).map((item, index) => ( // Duplicate menu items
      <div
        onClick={() =>
          setcategory((prev) =>
            prev === item.menu_name ? "All" : item.menu_name
          )
        }
        key={index}
        className="explore-menu-list-item"
      >
        <img
          className={category === item.menu_name ? "active" : ""}
          src={item.menu_image}
          alt={item.menu_name}
        />
        <p>{item.menu_name}</p>
      </div>
    )
  )}
  </div>
</div>

<hr />
</div>

);
};
```

export default ExploreMenu;

FoodDisplay.jsx

```
import React, { useContext, useState,
useEffect } from "react";
import "../FoodDisplay.css";
import { StoreContext } from
"../../Context/StoreContext";
```

```
const FoodDisplay = ({ category,
setshowlogin }) => {
  const { food_list = [], addToCart } =
useContext(StoreContext);
  const [selectedFood,
setSelectedFood] = useState(null);
  const [quantity, setQuantity] =
useState(1);
  const [cartMessage, setCartMessage]
= useState("");
  const [loggedInUser,
setLoggedInUser] = useState(null);

  useEffect(() => {
    const storedUser =
localStorage.getItem("user");
    setLoggedInUser(storedUser ?
JSON.parse(storedUser) : null);
  }, []);

  useEffect(() => {
    const handleStorageChange = () =>
{
      const storedUser =
localStorage.getItem("user");
      setLoggedInUser(storedUser ?
JSON.parse(storedUser) : null);
    };

    window.addEventListener("storage"
, handleStorageChange);
    return () =>
window.removeEventListener("storage
", handleStorageChange);
  }, []);

  const handleFoodClick = (item) => {
    setSelectedFood(item);
    setQuantity(1);
  };

  const handleAddToCart = () => {
    if (!loggedInUser) {
      setCartMessage("✗ Please log in
```

```
to add items to the cart!");
    setTimeout(() =>
setCartMessage(""), 3000);
    if (typeof setshowlogin ===
"function") setshowlogin(true);
    return;
}

if (selectedFood) {
    addToCart(selectedFood._id,
quantity);
    setCartMessage(` 
${selectedFood.name} added to cart!
 `);
    setTimeout(() =>
setCartMessage(""), 3000);
    setSelectedFood(null);
    setQuantity(1);
}
};

return (
<div className="food-display"
id="food-display">
    <h2>Food Menu ...!! Visit
Now...</h2>
    <div className="food-display-
list">
        {food_list
        .filter((item) => category ===
"All" || item.category === category)
        .map((item) => (
            <div key={item._id}
className="food-item" onClick={()
=> handleFoodClick(item)}>
                <img src={item.image}
alt={item.name} className="food-
image" />
                <p>{item.name}</p>
            </div>
        ))}
    </div>
```

```
{selectedFood && (  
  <div className="food-item-  
popup">  
    <div className="popup-  
content">  
      <span className="close-btn"  
onClick={() =>  
setSelectedFood(null)}>&times;</span  
>  
      <img  
src={selectedFood.image}  
alt={selectedFood.name}  
className="popup-image" />  
      <h2>{selectedFood.name}</h2  
>  
      <p>{selectedFood.description}  
</p>  
      <p>Price:  
Rs. {selectedFood.price}</p>  
  
      <div className="quantity-  
control">  
        <button onClick={() =>  
setQuantity(Math.max(1, quantity -  
1))}>-</button>  
        <span>{quantity}</span>  
        <button onClick={() =>  
setQuantity(quantity + 1)}>+</button>  
      </div>  
  
      <button className="add-to-  
cart-btn"  
onClick={handleAddToCart}>Add to  
Cart</button>  
    </div>  
  </div>  
)}  
  
{cartMessage && <div  
className="cart-  
message">{cartMessage}</div>  
</div>  
);  
};
```

```
export default FoodDisplay;
```

Footer.jsx

```
import React from "react";
import "./Footer.css";
import { assets } from
"../../assets/assets";
import { Link, useNavigate } from
"react-router-dom";

const Footer = () => {
  const navigation = useNavigate();
  return (
    <div className="footer"
id="footer">
      <div className="footer-content">
        <div className="image">
          <a href="#navbar">
            <img src={assets.logo}
alt="Logo" />
          </a>
        </div>

        <div className="footer-left">
          <ul>
            <u>
              <a href="#navbar">
                {" "}
                <li> 🏠 Home</li>
              </a>
            </u>
            <u>
              <a href="#explore-menu">
                {" "}
                <li> 📖 Menu</li>{" "}
              </a>
            </u>
            <u>
              <a href="#app-download">
                {" "}
                <li> 📱 Mobile App</li>
              </a>
```

</u>

<u>

{ " "

  Contact Us

</u>

<div className="footer-right">

<u>

{ " "

  Facebook

</u>

<u>

{ " "

  Twitter

</u>

<u>

{ " "

  LinkedIn{ " "

</u>

</div>

</div>

<div className="bus">

<u>

<Link to="/celebratie">

<p>  Business

Partners</p>

</Link>

</u>

</div>

<p> 

jkhungershub2514@gmail.com</p>

<p>  9943934520</p>

</div>

<div className="social-media">

<img

src={assets.facebook_icon}

alt="Facebook" />

<a

href="https://x.com/i/flow/login"

target="_blank">

<img src={assets.twitter_icon}

alt="Twitter" />

<a

href="https://www.linkedin.com/login/"

target="_blank">

<img src={assets.linkedin_icon}

alt="LinkedIn" />

</div>

<div className="copy-right">

<p>

© 2025 — <span

className="brand">  JK HUNGER

HUB. All rights

reserved.

</p>

</div>

</div>

);

};

export default Footer;

6. Navbar.jsx

```
import React, { useContext, useState } from 'react'
import './Navbar.css'
import { assets } from '../assets/assets'
import { Link } from 'react-router-dom';
import { StoreContext } from '../Context/StoreContext';

const Navbar = ({setshowlogin}) => {
  const [menu,setmenu]=useState("Home");
  const {getTotalAmount}=useContext(StoreContext);
  return (
    <div className='navbar' id='navbar'>
      <Link to='/'><img src={assets.logo} alt="" className="logo" /></Link>
      <ul className="navbar-menu">
        <a href='#explore-menu' onClick={()=>setmenu("Home")} className={menu==="Home"? "active":""}> Home
        | </a>
        <a href='#food-display' onClick={()=>setmenu("Menu")} className={menu==="Menu"? "active":""}> Menu |
        </a>
        <a href='#app-download' onClick={()=>setmenu("Mobile-App")} className={menu==="Mobile-
        App"? "active":""}> Mobile-App | </a>
        <a href='#footer' onClick={()=>setmenu("Contact Us")} className={menu==="Contact Us"? "active":""}>
        Contact Us | </a>
      </ul>
      <div className="navbar-right">
        <img src={assets.search_icon} alt="" />
        <div className="navbar-search-icon">
          <Link to='/cart'> <img src={assets.basket_icon} alt="" /></Link>
          <div className={getTotalAmount()===0? "<>/>:" : "dot"}></div>
        </div>
        <button onClick={()=>setshowlogin(true)}>Log in </button>
      </div>
    </div>
  )
}
```

export default Navbar

7. Results & Screenshots

Key Results

- ☐ **Authentication with email/password and Google OAuth**
- ☐ **Food item browsing by category with add-to-cart functionality**
- ☐ **Cart and checkout system integrated with order placement**
- ☐ **Order tracking with status updates (pending, confirmed, delivered)**
- ☐ **Admin panel for food item management (add/update/delete)**
- ☐ **Promotional/celebration pages for offers and deals**
- ☐ **Responsive navigation (Navbar, Header, Footer) across devices**
- ☐ **Contact form functional with backend integration**

SCREENSHORTS



[Home](#) | [Menu](#) | [Mobile-App](#) | [Contact Us](#)



[Log in](#)



Explore Your Favourite Dish with Us

Explore a world of flavors with our food ordering app! From sizzling starters to indulgent desserts, our curated menu has something for everyone. Choose from fresh salads, hearty mains, and international cuisines, all crafted to perfection. Craving comfort food or trying something new? Dive into our mouthwatering selection now!



Rolls



Desserts



Sandwich



Cake



Pure Veg



Pasta



Noodle

Food Menu ...!! Visit Now...



Greek salad



Veg salad



Clover Salad



Chicken Salad



Buttter Noodles



Veg Noodles



Somen Noodles



Cooked Noodles

Download app for better experience



[Home](#)
[Menu](#)
[Mobile App](#)
[Contact Us](#)

[Facebook](#)
[Twitter](#)
[LinkedIn](#)

Business Partners

jkhungerhub2514@gmail.com

9943934520



© 2025 — JK HUNGER HUB. All rights reserved.

Log in

Log In

Create a new account [Click here](#)

	Jar Ice Cream	Rs. 100.00	1	Rs. 100.00	
	Vegan Sandwich	Rs. 70.00	1	Rs. 70.00	

Cart Totals

SubtotalRs. 420.00

Delivery FeeRs. 40.00

TotalRs. 460.00

PROCEED TO CHECKOUT



Delivery Information

Cart Totals

Subtotal420

Delivery Fee40

Total460

Place Order



[Home](#) | [Menu](#) | [Mobile-App](#) | [Contact Us](#)



PARTNERS



Mr.JAI
JK HUNGER HUB



Mr.DEEPAK
RD GROCERY SHOP



Mr.XAVIER
DX CAFE



Mr.ASHWANTH
STAR RENTAL



Mr.HARIPRASATH
BOOK LAND



Mr.AADHITHIYA
KSA HOSPITAL

8. References

- **React Router Documentation:** <https://reactrouter.com/>
- **Material-UI (MUI):** <https://mui.com/>
- **Chart.js:** <https://www.chartjs.org/>
- **Axios:** <https://axios-http.com/>
- **Animate.css:** <https://animate.style/>
- **Node.js Documentation:** <https://nodejs.org/en/docs>
- **Express.js:** <https://expressjs.com/>
- **MongoDB Atlas:** <https://www.mongodb.com/atlas>

Appendix

A. Directory Structure

```
mern_frontend/  
├── dist/  
├── node_modules/  
├── public/  
├── src/  
│   ├── assets/  
│   ├── components/  
│   │   ├── AppDownload/  
│   │   │   ├── AppDownload.css  
│   │   │   └── AppDownload.jsx  
│   │   ├── ExploreMenu/  
│   │   │   ├── ExploreMenu.css  
│   │   │   └── ExploreMenu.jsx  
│   │   ├── FoodDisplay/  
│   │   │   ├── FoodDisplay.css  
│   │   │   └── FoodDisplay.jsx  
│   │   ├── FoodItem/  
│   │   │   ├── FoodItem.css  
│   │   │   └── FoodItem.jsx  
│   │   ├── Footer/  
│   │   │   ├── Footer.css  
│   │   │   └── Footer.jsx  
│   │   ├── Header/  
│   │   │   ├── Header.css  
│   │   │   └── Header.jsx  
│   │   └── LoginPop/  
│   │       ├── LoginPop.css  
│   │       └── LoginPop.jsx
```

